Design Baseline

Named input case · design_baseline.json

2026-10-31T14:36:14+00:00

Dry rover mass	Payload per cycle	Complete cycles	Delivered volume
43.9 kg	**103.3 kg**	**3**	**185.9 L**
Before adding regolith	Total carried by four drums	Within 15 minutes; no partial delivery credit	Complete-cycle mission estimate

Construction score

Continuous mission estimate using the configured mass and energy weights. The complete-cycle estimate is reported in Performance & mass.

Construction points	Mass points	Energy points
1,939.74	**1,585.00**	**354.74**
Weighted mass + energy points	Dry-mass productivity component	Consumed-energy productivity component

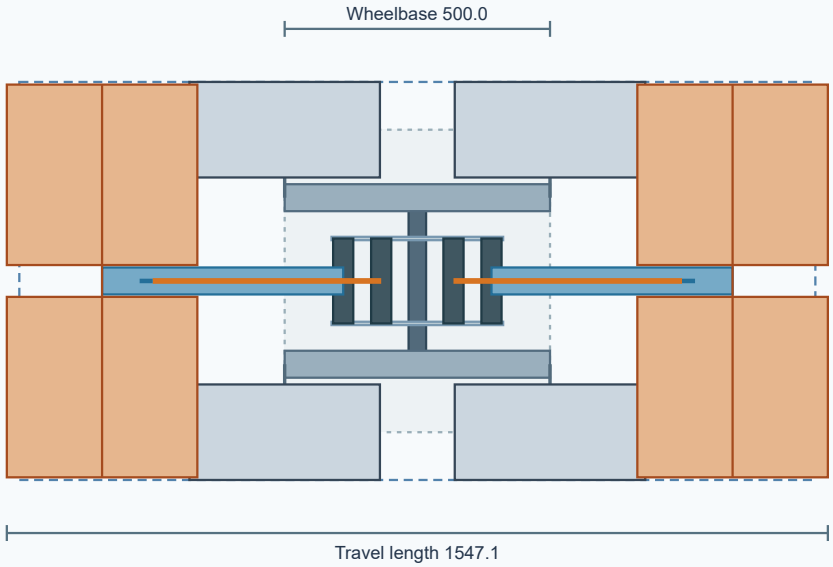
01 / ROVER CONFIGURATION

Geometry at the evaluated inputs

Level-ground schematics and one prescribed obstacle pose. Check statuses below determine whether envelopes and clearance requirements are satisfied.

Top-down view / travel pose

Orthographic footprints; front is +x (right). Dimensions in mm.



Wheel-center track: 570.0 | Overall width: 750.0 | Wheel width: 180.0

Each drum: 359.0 diameter x 340.0 axial length | Pair gap: 60.0

Blue: one arm per drum pair · orange: actuators · charcoal: aluminum cross-tubes · slate: rockers and axle.

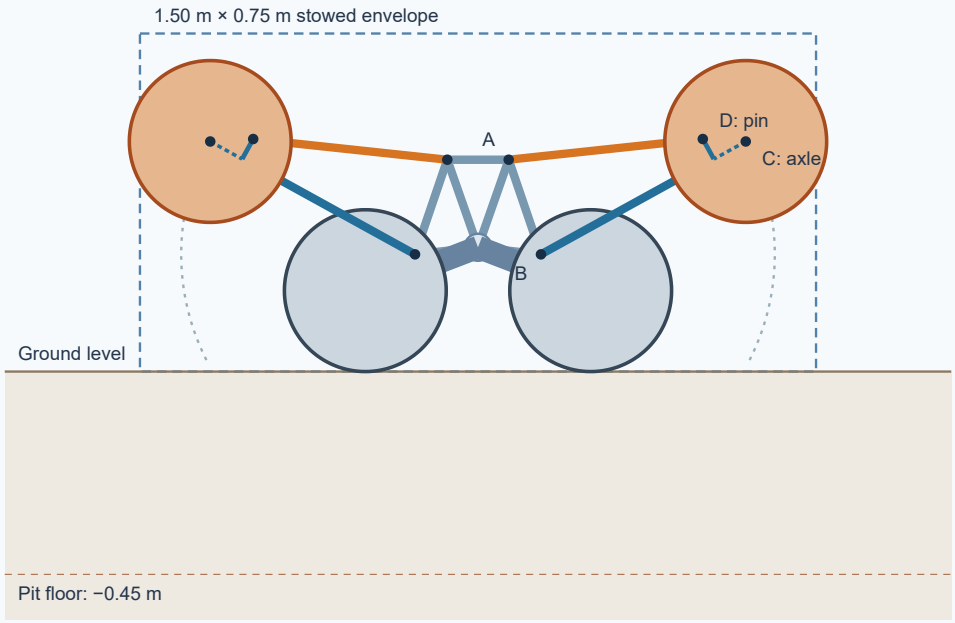
Four aluminum cross-tubes: upper/lower footprints can overlap. Dashed blue: 1500 x 750 stowed limit.

Plate outlines are schematic. Projection overlap does not prove collision. Heights and equipment are not shown.

Travel-pose plan view. Four wheels and four drums are shown at their model stations. Top-down SVG

Travel

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



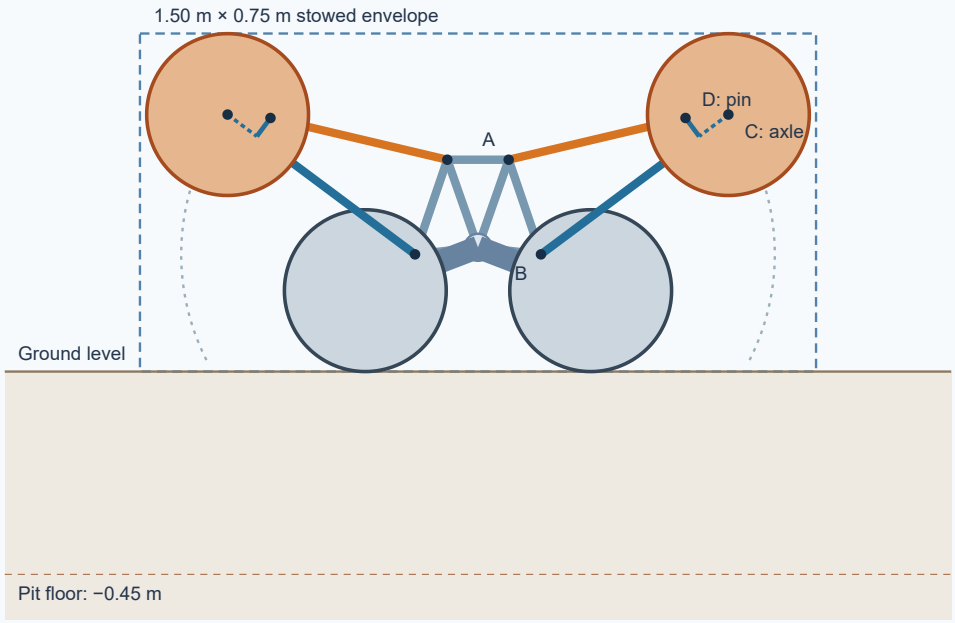
Actuator: 433.1 mm · Fixed arm: 518.8 mm · Drum Ø 359.0 mm

Drum top: +0.690 m · Drum bottom: +0.331 m · A: 0.470 m · B: 0.260 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -80.8 mm / top +50.8 mm · Ground is y = 0

Deposition · both pairs

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



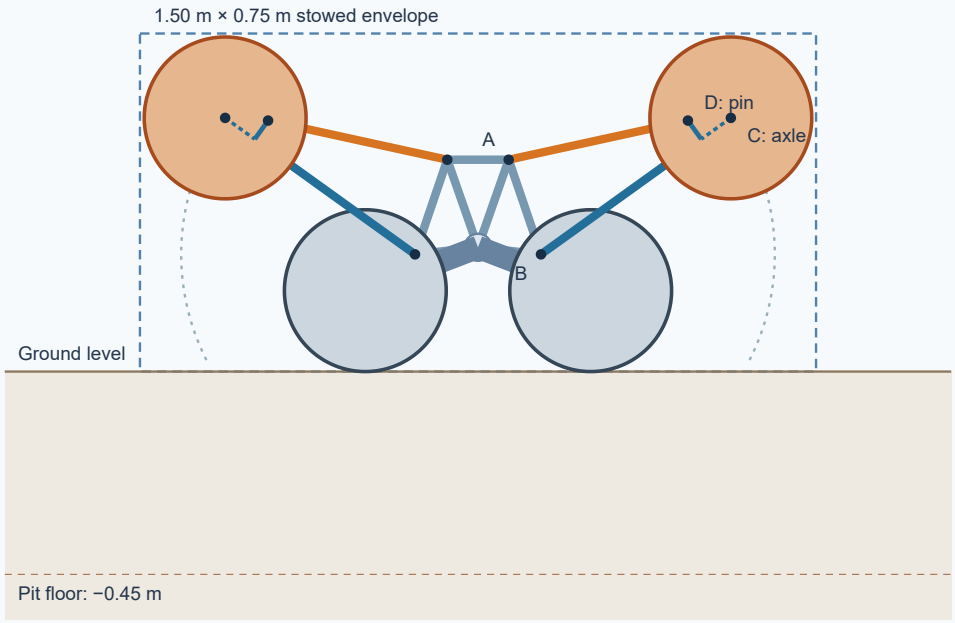
Actuator: 403.1 mm · Fixed arm: 518.8 mm · Drum Ø 359.0 mm

Drum top: +0.750 m · Drum bottom: +0.391 m · A: 0.470 m · B: 0.260 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -80.8 mm / top +50.8 mm · Ground is y = 0

Compact · starting envelope

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



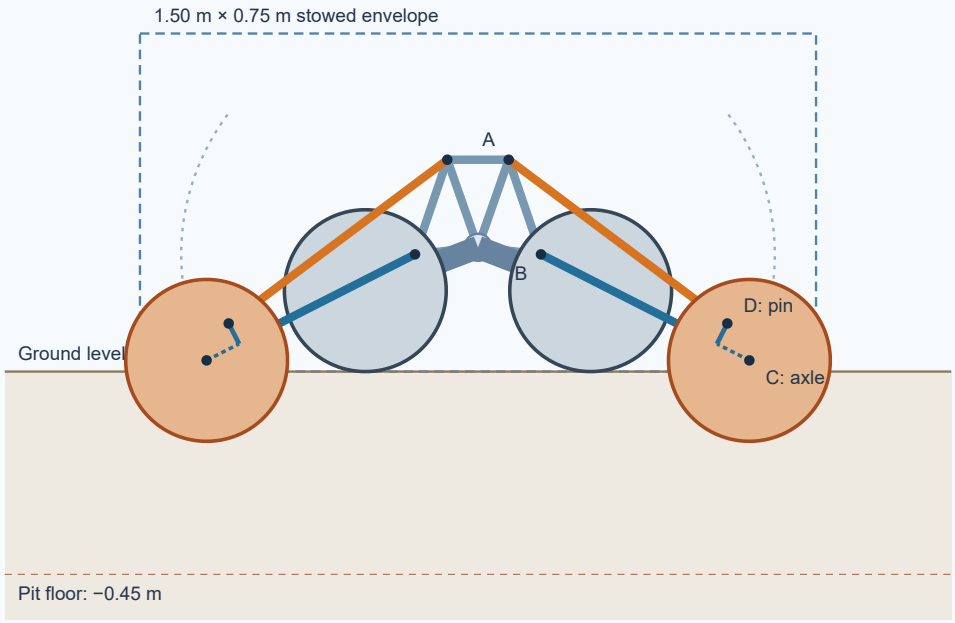
Actuator: 407.0 mm · Fixed arm: 518.8 mm · Drum Ø 359.0 mm

Drum top: +0.742 m · Drum bottom: +0.383 m · A: 0.470 m · B: 0.260 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -80.8 mm / top +50.8 mm · Ground is y = 0

Maximum extension · plunge / recovery reach

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



Actuator: 606.0 mm · Fixed arm: 518.8 mm · Drum Ø 359.0 mm

Drum top: +0.204 m · Drum bottom: -0.155 m · A: 0.470 m · B: 0.260 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -80.8 mm / top +50.8 mm · Ground is y = 0

Blue: arm · orange: actuator. Each visible drum represents a shaft-linked pair. Drawings are not CAD.

Travel and deposition mass properties

Pose / loading	Longitudinal CG	CG height	Yaw inertia
Travel · fully loaded	-2.41e-14 mm	425.1 mm	50.9 kg·m²
Deposition · fully loaded	-2.41e-14 mm	473.7 mm	45.6 kg·m²
Deposition turn · one pair full	300.3 mm	453.9 mm	26.7 kg·m²
Deposition · empty	-2.53e-14 mm	387.5 mm	7.77 kg·m²

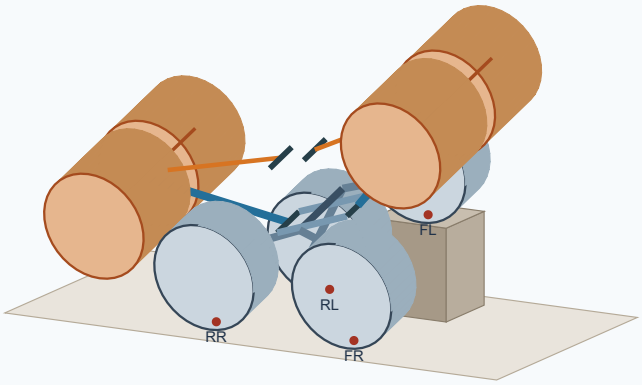
Added pose-change duration: 5.84 s. Added electrical energy: 1,709 J. Both pairs move together while full before dumping, then return empty after both dumps.

Raising reduces yaw inertia only when the drums move inward. Higher CG can reduce dynamic stability. One wheelbase must satisfy both travel and deposition checks. Half-loaded arm repositioning and dynamic dumping require controller and hardware validation.

One wheel on the rock / travel pose

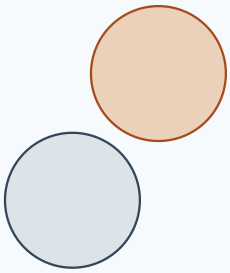
Front-left raised: 300.0 mm · chassis pitch 19.0° · roll 13.2°

Same kinematic pose as the model; arms held in travel position.



Common roll removed

Closest cylinder pair: drum 1 / FL



Radial circle gap: +48.60 mm

Axial end gap: -175.00 mm

Finite-cylinder gap: +48.60 mm

Required clearance: 15.00 mm

One centered arm/actuator per drum pair · blue: arms · orange: actuators · charcoal: aluminum cross-tubes · gray: wheels · tan: drums.

Support block shows the prescribed height; its shape and width are illustrative. Actual rock climbing is not simulated.

Contact model omits camber effects. The right section also needs the axial-width check; circles alone can over-reject.

Parallel-cylinder clearance is the default model. This section shows the closest pair for the illustrated front-left raise.

Plate-frame member schematic

Plate frame: provisional member topology

Lengths and directions follow the mounting coordinates; this is not the manufactured cutout outline.

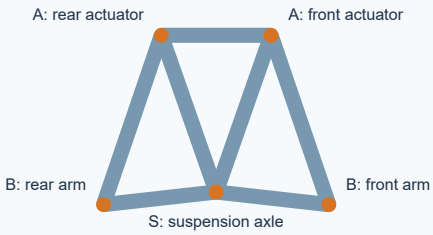


Plate thickness: 6.35 mm (fixed)

Common strip width: 18.75 mm

cross-tube ID / OD: 31.75 / 38.00 mm

Plate spacing / cross-tube span: 160.0 mm

Rocker bearing spacing: 313.8 mm

Wheel-to-rocker gap: 12.7 mm

Aluminum arm box: 50.80 x 50.80 mm

Arm wall: 1.5875 mm

Central support: pin, balanced mirrored gravity only.

cross-tube material properties are PROVISIONAL.

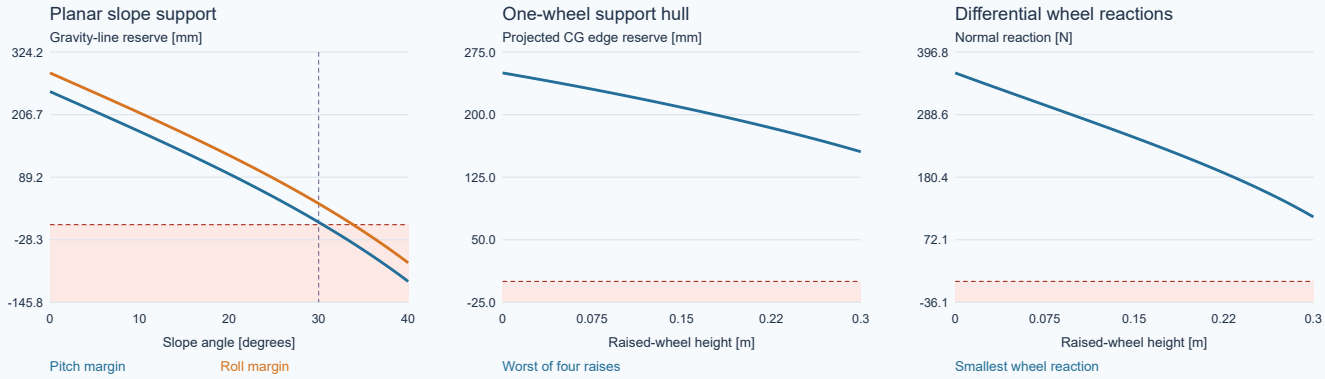
Spoke buckling is a strip screen,

not whole-plate buckling validation.

Static slope and articulation curves

Loaded travel pose: gravity stability screens

One wheel raised at a time on horizontal platforms. Curves are model calculations; no physical rock profile is simulated.



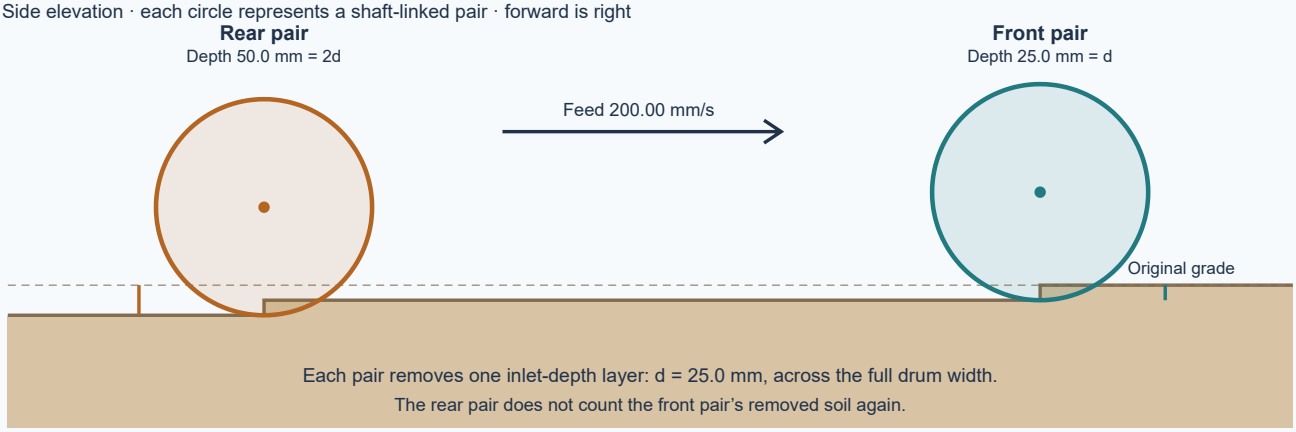
EXCAVATION / FORWARD FEED

Drive straight with the drums lowered

Each pair cuts one inlet-depth fresh layer across its full drum width. Radial protrusion equals inlet depth; the rear cuts a second layer in the same tracks. Forward advance per scoop remains limited by inlet depth.

Moving excavation · two fresh layers

Side elevation · each circle represents a shaft-linked pair · forward is right



Staged entry: front starts → rear reaches that start and lowers → front lifts when full → rear finishes.

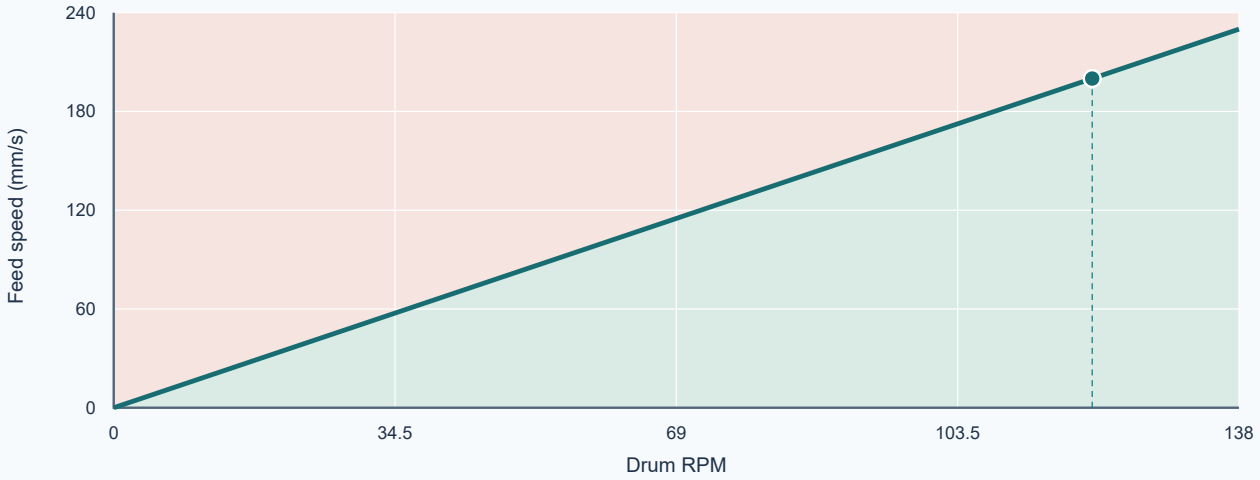
Productive lane: 3.115 m · total feed travel: 4.404 m · feed time: 23.5 s

The maximum-extension view is a separate plunge/recovery reach screen; it is not this operating pose.

Schematic soil boundary; entry/endcap soil is not credited. Soil flow and recovery driving require validation.

Feed speed and drum RPM

Forward advance per scoop ≤ inlet depth · an uncalibrated feed screen



Current: 120.00 rpm, 200.00 mm/s

Forward advance: 25.00 mm/scoop · inlet depth: 25.00 mm

Maximum feed = inlet_depth × scoops/revolution × RPM / 60; at this RPM: 200.00 mm/s.

Green: passes this feed screen. Red: excessive advance per scoop. Motor, reach and traction checks still apply.

This volume/bite constraint does not establish measured capture, cutting loads or high-speed soil retention.

Excavation quantity	Evaluated value
Inlet depth / fresh-layer cut depth	25.0 mm
Inlet width (full drum width)	340.0 mm
Front depth below original grade	25.0 mm
Rear depth below original grade	50.0 mm
Straight excavation feed speed	200.0 mm/s
Excavation drum RPM	120.0 rpm
Advance per scoop	25.0 mm
Feed limit at selected drum RPM	200.0 mm/s
Productive excavation lane length	3,115 mm
Feed travel including tandem staging	4,404 mm
Straight-feed duration	23.5 s
Feed-drive electrical energy	3,077 J
Feed-drive peak electrical power	210.4 W

Excavation quantity	Evaluated value
Plunge/recovery: axle reaches ground level	-24.8 mm

Feed-drive electrical work and peak power are included in digging totals. The old stationary-plunge soil-volume calculation is inactive. Maximum extension remains a separate capability envelope. Reaching the ground with an axle does not verify motor clearance or driving on the drums.

02 / PREDICTED OUTPUT

Performance & mass

Complete-cycle estimates and continuous MDO estimates are labeled separately.

Prediction	Value	Meaning
Full cycle duration	235.2 s	Includes digging, actuation, travel, dumping and relocation; overlap removed once.
Energy per cycle	17.5 Wh	Electrical energy including auxiliaries.
Mission energy · complete cycles	53.7 Wh	Includes idle electronics for remaining mission time.
Mission energy · continuous estimate	66.8 Wh	Used by the existing battery constraint and MDO objective.
Peak electrical power	3,037 W	Largest modeled simultaneous demand; not an average power rating.
Loaded mass	147.1 kg	Dry mass + all four drums' regolith payload.
Traversal timing mode	fixed	Fixed schedule
Loaded outbound route	45.0 s	Includes route turns; excludes deposition and relocation.
Empty return route	45.0 s	Includes route turns; excludes deposition and relocation.
Actual route peak speed	0.57 m/s	Prescribed motion, not measured vehicle performance.
Score · complete-cycle estimate	1,588.44 points	Uses complete deliveries and idle remainder; separate from the continuous MDO objective.
Empty rover · 30° climb	No climbing solution	Force shortfall per wheel: 32.02 N. Acceptance: solved slip strictly below 1.
Full rover · 20° climb	No climbing solution	Force shortfall per wheel: 40.88 N. Acceptance: solved slip strictly below 1.

Dry mass breakdown

Each assembly is counted once.

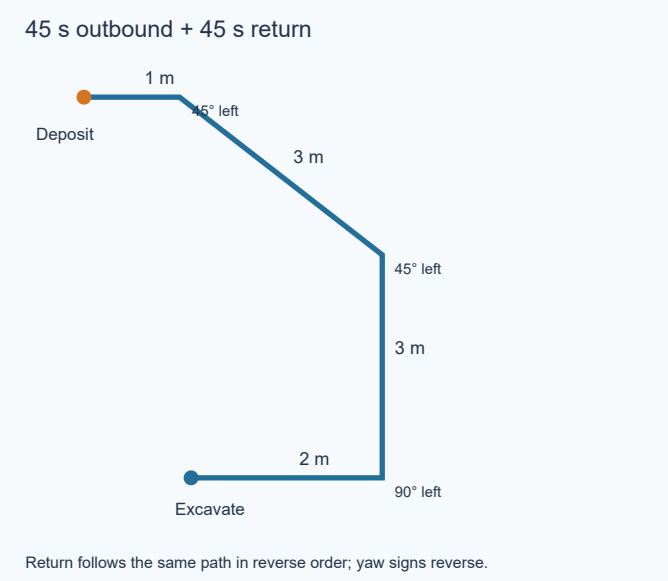
Equipment allowance	6.00 kg
Frame	5.50 kg
Four bare wheels	6.49 kg
Four drive assemblies	4.15 kg
Four drums + two drive assemblies	13.9 kg
Battery	1.85 kg
Arms + actuators	5.99 kg

Electrical energy per cycle

Work / provisional efficiency, plus auxiliaries.

Digging	10.8 Wh
Dumping	0.14 Wh
Travel + deposition turn	1.64 Wh
Excavation actuators	2.74 Wh
Deposition pose changes	0.47 Wh
Fresh-site move	0.07 Wh
Electronics	1.63 Wh

Route & cycle accounting



45 s outbound + 45 s return

Position each pair → drive the staged straight excavation pass → raise to travel → outbound → move to deposition pose → dump one pair → turn → dump second pair → return empty drums to travel pose → return route → relocate to a fresh lane.

Moving excavation has no lowering/cutting overlap credit. Full-range arm motion and load allowances conservatively screen the plunge/recovery sweep. The deposition turn and fresh-site relocation are additional to the timed route. Modeled full cycle: 235.2 s.

03 / GAPS TO RESOLVE

Requirements & model checks

Failures are listed first. “Not assessed” means the model does not verify that capability.

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Empty rover: 30° climb `empty_slope_traversal_margin`	No climbing solution `Slip = 1 is a failure sentinel`	Slip < 1 (strict)	-0.34	Fail
Full rover: 20° climb `loaded_slope_traversal_margin`	No climbing solution `Slip = 1 is a failure sentinel`	Slip < 1 (strict)	-0.12	Fail
Excavation shaft-power reserve `drum_power_margin`	Reserve: -817.2 W	Reserve ≥ 0	-817.2 W	Fail
Excavation: traction reserve `dig_traction_margin`	Reserve: -654.0 N	Reserve ≥ 0	-654.0 N	Fail
Turning: traction reserve `turn_traction_margin`	Reserve: -250.4 N	Reserve ≥ 0	-250.4 N	Fail
Battery peak-power capacity `battery_power_margin`	3,037 W	≤ 1,728 W	-1,309 W	Fail
Travel drum height above wheel-diameter step `wheel_first_margin`	Reserve: -28.0 mm	Reserve ≥ 0	-28.0 mm	Fail
Startup position margin `startup_position_margin`	Reserve: -26.1 mm	Reserve ≥ 0	-26.1 mm	Fail
Excavation torque reserve `drum_torque_margin`	Reserve: -143.0 N·m	Reserve ≥ 0	-143.0 N·m	Fail
Asymmetric dump tip reserve `tip_stability_margin`	Reserve: -50.3 mm	Reserve ≥ 0	-50.3 mm	Fail
Available excavation area `excavation_area_margin`	Reserve: -1.49 m^2	Reserve ≥ 0	-1.49 m^2	Fail
Plunge/recovery: axle reaches ground level `axle_ground_margin`	24.8 mm	≤ 0.00 mm	-24.8 mm	Fail
Excavation feed: traction reserve `feed_traction_margin`	Reserve: -654.0 N	Reserve ≥ 0	-654.0 N	Fail

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Dry rover mass `dry_mass`	43.9 kg	≤ 80.0 kg	36.1 kg	Pass
Compact overall length `rover_length`	1,481 mm	≤ 1,500 mm	19.0 mm	Pass
Overall width `rover_width`	750.0 mm	≤ 750.0 mm	0.00 mm	Pass
Compact overall height `rover_height`	742.3 mm	≤ 750.0 mm	7.72 mm	Pass
Ground clearance: lower axle after wheel/bar deductions `ground_clearance_margin`	Reserve: 27.3 mm	Reserve ≥ 0	27.3 mm	Pass
Excavation: penetration reserve `penetration_margin`	Reserve: 124.3 N	Reserve ≥ 0	124.3 N	Pass
Inlet depth: radial geometry reserve `cut_depth_margin`	Reserve: 19.9 mm	Reserve ≥ 0	19.9 mm	Pass
Ground contact pressure `ground_pressure_margin`	18.8 kPa	≤ 25.0 kPa	6.24 kPa	Pass
Straight travel: motor-force reserve `straight_motor_margin`	Reserve: 3,545 N	Reserve ≥ 0	3,545 N	Pass
Turning: motor-force reserve `turn_motor_margin`	Reserve: 5,327 N	Reserve ≥ 0	5,327 N	Pass
Straight travel: traction reserve `straight_traction_margin`	Reserve: 201.1 N	Reserve ≥ 0	201.1 N	Pass
Straight travel: shaft-power reserve `straight_power_margin`	Reserve: 2,495 W	Reserve ≥ 0	2,495 W	Pass
Turning: shaft-power reserve `turn_power_margin`	Reserve: 2,587 W	Reserve ≥ 0	2,587 W	Pass
Battery usable energy `battery_energy_margin`	66.8 Wh	≤ 207.4 Wh	140.5 Wh	Pass
Time remaining after one complete cycle `complete_cycle_margin`	Reserve: 664.8 s	Reserve ≥ 0	664.8 s	Pass
Pit depth margin `pit_depth_margin`	Reserve: 295.3 mm	Reserve ≥ 0	295.3 mm	Pass
Transport clearance margin `transport_clearance_margin`	Reserve: 303.3 mm	Reserve ≥ 0	303.3 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Deposition pose: triangle reach reserve `deposition_reach_margin`	Reserve: 169.0 mm	Reserve ≥ 0	169.0 mm	Pass
Deposition command below maximum extension `deposition_position_margin`	Reserve: 202.9 mm	Reserve ≥ 0	202.9 mm	Pass
Deposition drum above ground `deposition_clearance_margin`	Reserve: 310.7 mm	Reserve ≥ 0	310.7 mm	Pass
Dig position margin `dig_position_margin`	Reserve: 199.0 mm	Reserve ≥ 0	199.0 mm	Pass
Pivot spacing margin `pivot_spacing_margin`	Reserve: 141.9 mm	Reserve ≥ 0	141.9 mm	Pass
Pivot mount margin `pivot_mount_margin`	Reserve: 239.0 mm	Reserve ≥ 0	239.0 mm	Pass
Pivot longitudinal margin `pivot_longitudinal_margin`	Reserve: 110.3 mm	Reserve ≥ 0	110.3 mm	Pass
Actuator pivot longitudinal margin `actuator_pivot_longitudinal_margin`	Reserve: 181.9 mm	Reserve ≥ 0	181.9 mm	Pass
Bottom mount margin `bottom_mount_margin`	Reserve: 80.5 mm	Reserve ≥ 0	80.5 mm	Pass
Stroke margin `stroke_margin`	Reserve: 132.9 mm	Reserve ≥ 0	132.9 mm	Pass
Triangle retracted margin `triangle_retracted_margin`	Reserve: 199.0 mm	Reserve ≥ 0	199.0 mm	Pass
Triangle extended margin `triangle_extended_margin`	Reserve: 41.8 mm	Reserve ≥ 0	41.8 mm	Pass
Toggle margin `toggle_margin`	Reserve: 400.9 mm	Reserve ≥ 0	400.9 mm	Pass
Actuator toggle margin `actuator_toggle_margin`	Reserve: 326.8 mm	Reserve ≥ 0	326.8 mm	Pass
Wheel clearance margin `wheel_clearance_margin`	Reserve: 7.00 mm	Reserve ≥ 0	7.00 mm	Pass
Deployed height margin `deployed_height_margin`	Reserve: 1,750 mm	Reserve ≥ 0	1,750 mm	Pass
Actuator force capacity `actuator_force_margin`	2,500 N	≤ 4,003 N	1,503 N	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Dumping torque reserve `dump_torque_margin`	Reserve: 88.8 N·m	Reserve ≥ 0	88.8 N·m	Pass
Minimum payload reserve `payload_margin`	Reserve: 103.2 kg	Reserve ≥ 0	103.2 kg	Pass
Excavated trench diameter reserve `trench_traverse_margin`	Reserve: 309.0 mm	Reserve ≥ 0	309.0 mm	Pass
Fresh site margin `fresh_site_margin`	Reserve: 60.0 mm	Reserve ≥ 0	60.0 mm	Pass
Frame section margin `frame_section_margin`	Reserve: 7.25 mm	Reserve ≥ 0	7.25 mm	Pass
Arm section margin `arm_section_margin`	Reserve: 11.1 mm	Reserve ≥ 0	11.1 mm	Pass
Structural stress envelope `structure_stress_margin`	98.2 MPa	≤ 120.0 MPa	21.8 MPa	Pass
Structural deflection envelope `structure_deflection_margin`	0.55 mm	≤ 5.00 mm	4.45 mm	Pass
Structural buckling reserve `structure_buckling_margin`	Reserve: 2,352 N	Reserve ≥ 0	2,352 N	Pass
Level-route dynamic tip reserve `mobility_stability_margin`	Reserve: 244.3 mm	Reserve ≥ 0	244.3 mm	Pass
Rock pose: geometric reach reserve `obstacle_pose_margin`	Reserve: 182.7 mm	Reserve ≥ 0	182.7 mm	Pass
Rock endpoint: global tip reserve `obstacle_tip_margin`	Reserve: 155.5 mm	Reserve ≥ 0	155.5 mm	Pass
Obstacle drum–wheel clearance `obstacle_drum_wheel_margin`	48.6 mm	≥ 15.0 mm	33.6 mm	Pass
30° slope: static tip reserve `slope_tip_margin`	Reserve: 4.59 mm	Reserve ≥ 0	4.59 mm	Pass
Rock path: global tip reserve `rock_sweep_tip_margin`	Reserve: 155.5 mm	Reserve ≥ 0	155.5 mm	Pass
Rock path: smallest wheel reaction `rock_sweep_wheel_load_margin`	Reserve: 111.6 N	Reserve ≥ 0	111.6 N	Pass
Fresh-site move: drive/traction reserve `site_drive_margin`	Reserve: 172.9 N	Reserve ≥ 0	172.9 N	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Plate geometry reserve `plate_geometry_margin`	Reserve: 33.9 mm	Reserve ≥ 0	33.9 mm	Pass
Frame packaging reserve `frame_packing_margin`	Reserve: 63.7 mm	Reserve ≥ 0	63.7 mm	Pass
Plate strip stress `plate_stress_margin`	12.3 MPa	≤ 120.0 MPa	107.7 MPa	Pass
Plate strip buckling reserve `plate_buckling_margin`	Reserve: 2,352 N	Reserve ≥ 0	2,352 N	Pass
aluminum cross-tube bending stress `cross_tube_bending_margin`	46.3 MPa	≤ 120.0 MPa	73.7 MPa	Pass
Actuator catalog stroke domain `actuator_stroke_domain_margin`	Reserve: 152.4 mm	Reserve ≥ 0	152.4 mm	Pass
Actuator load-curve domain `actuator_curve_margin`	Reserve: 1,503 N	Reserve ≥ 0	1,503 N	Pass
Forward advance per scoop: inlet-depth reserve `feed_margin`	25.0 mm	≤ 25.0 mm	0.00 mm	Pass
Moving excavation: operating-pose reach `operating_reach_margin`	Reserve: 30.0 mm	Reserve ≥ 0	30.0 mm	Pass
Moving excavation: geometry domain `excavation_domain_margin`	Reserve: 154.5 mm	Reserve ≥ 0	154.5 mm	Pass
Excavation feed: motor-force reserve `feed_motor_margin`	Reserve: 5,599 N	Reserve ≥ 0	5,599 N	Pass
Excavation feed: shaft-power reserve `feed_power_margin`	Reserve: 2,528 W	Reserve ≥ 0	2,528 W	Pass
Lin act side arm offset search range `lin_act_side_arm_offset_bounds`	-80.8 mm	-300.0 mm to -50.8 mm	30.0 mm	Pass
Lin act top arm offset search range `lin_act_top_arm_offset_bounds`	50.8 mm	-150.0 mm to 150.0 mm	99.2 mm	Pass
Wheel diameter search range `wheel_diameter_bounds`	359.0 mm	180.0 mm to 359.0 mm	0.00 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Wheel width search range `wheel_width_bounds`	180.0 mm	50.0 mm to 359.0 mm	130.0 mm	Pass
Drum outer diameter search range `drum_diameter_bounds`	359.0 mm	160.0 mm to 359.0 mm	0.00 mm	Pass
Width of each drum search range `drum_length_bounds`	340.0 mm	120.0 mm to 359.0 mm	19.0 mm	Pass
Excavation drum RPM search range `drum_rpm_bounds`	120.0 rpm	5.00 rpm to 120.0 rpm	0.00 rpm	Pass
Straight excavation feed speed search range `feed_speed_bounds`	0.20 m/s	0.0001 m/s to 0.50 m/s	0.20 m/s	Pass
Dumping speed search range `dump_rpm_bounds`	60.0 rpm	5.00 rpm to 60.0 rpm	0.00 rpm	Pass
Drum reduction ratio search range `drum_gear_ratio_bounds`	60.0	9.00 to 125.0	51.0	Pass
Inlet depth / fresh-layer cut depth search range `inlet_depth_bounds`	25.0 mm	3.00 mm to 25.0 mm	0.00 mm	Pass
Arm wall thickness search range `arm_wall_bounds`	1.59 mm	1.00 mm to 5.00 mm	0.59 mm	Pass
Battery capacity search range `battery_capacity_bounds`	259.2 Wh	10.0 Wh to 300.0 Wh	40.8 Wh	Pass
Travel actuator length search range `travel_actuator_length_bounds`	433.1 mm	403.1 mm to 606.3 mm	30.0 mm	Pass
Deposition actuator length search range `deposition_actuator_length_bounds`	403.1 mm	403.1 mm to 606.3 mm	0.00 mm	Pass
Compact actuator length search range `actuator_retracted_length_bounds`	407.0 mm	403.1 mm to 606.3 mm	3.90 mm	Pass
Maximum-extension actuator length search range `actuator_extended_length_bounds`	606.0 mm	403.1 mm to 606.3 mm	0.30 mm	Pass
Actuator pivot height search range `top_pivot_height_bounds`	470.0 mm	350.0 mm to 700.0 mm	120.0 mm	Pass
Actuator pivot longitudinal position search range `actuator_pivot_x_bounds`	68.1 mm	20.0 mm to 300.0 mm	48.1 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Arm pivot height search range `bottom_pivot_height_bounds`	260.0 mm	90.0 mm to 300.0 mm	40.0 mm	Pass
Arm length search range `arm_length_bounds`	518.8 mm	300.0 mm to 700.0 mm	181.2 mm	Pass
Arm pivot longitudinal position search range `pivot_x_bounds`	139.7 mm	50.0 mm to 400.0 mm	89.7 mm	Pass
Wheelbase search range `wheelbase_bounds`	500.0 mm	380.0 mm to 650.0 mm	120.0 mm	Pass
Wheel-center track search range `track_width_bounds`	570.0 mm	500.0 mm to 699.0 mm	70.0 mm	Pass
Aluminum arm box width search range `arm_box_width_bounds`	50.8 mm	25.0 mm to 100.0 mm	25.8 mm	Pass
Aluminum arm box height search range `arm_box_height_bounds`	50.8 mm	25.0 mm to 100.0 mm	25.8 mm	Pass
aluminum cross-tube span / plate spacing search range `cross_tube_span_bounds`	160.0 mm	160.0 mm to 460.0 mm	0.00 mm	Pass
Aluminum cross-tube wall thickness search range `cross_tube_wall_bounds`	3.12 mm	1.00 mm to 6.00 mm	2.12 mm	Pass
Plate strip width search range `plate_spoke_width_bounds`	18.8 mm	12.0 mm to 70.0 mm	6.75 mm	Pass
Rocker box width search range `rocker_box_width_bounds`	50.8 mm	20.0 mm to 65.0 mm	14.2 mm	Pass
Rocker box height search range `rocker_box_height_bounds`	50.8 mm	25.0 mm to 90.0 mm	25.8 mm	Pass
Rocker wall search range `rocker_wall_bounds`	1.59 mm	1.00 mm to 5.00 mm	0.59 mm	Pass
Axle outer diameter search range `axle_outer_diameter_bounds`	33.0 mm	25.0 mm to 90.0 mm	8.00 mm	Pass
Axle wall search range `axle_wall_bounds`	1.00 mm	1.00 mm to 6.00 mm	0.00 mm	Pass
Rocker pivot above wheel centers search range `rocker_pivot_offset_bounds`	96.2 mm	20.0 mm to 150.0 mm	53.8 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Actuator catalog stroke (continuous) search range `actuator_catalog_stroke_bounds`	203.2 mm	50.8 mm to 1,016 mm	152.4 mm	Pass
Drum-supported recovery driving	Axle-to-ground reach is screened geometrically. Drum traction, motor-ground clearance, recovery loads and actual escape capability require hardware and soil tests.	—	—	Not assessed
Slope motor limits / sideways sliding	The separate climbing checks accept slip<1 only. Motor capability on slopes, lateral sliding and controller behavior are not evaluated.	—	—	Not assessed
Real 30 cm rock traversal	Static poses and clearance screens do not establish the driving forces or actual terrain-contact sequence.	—	—	Not assessed
50 cm ditch crossing	Ditch width/radius and contact sequence have not been defined.	—	—	Not assessed
Continuous motor operation	Stator-current limits and thermal duty remain undecided.	—	—	Not assessed
Connections & material validation	Bonded inserts, bushings, bolts, local failure and selected aluminum cross-tube properties need separate checks.	—	—	Not assessed
Soil and skid-steer predictions	Rolling, traction, capture and scrub relationships remain uncalibrated.	—	—	Not assessed

Positive margins satisfy the modeled criterion. Status uses the saved model's numerical tolerance. Search ranges are hidden by default and available in the filter; they remain part of saved feasibility. Margins use their own units and cannot be ranked directly across unlike checks.

04 / ENGINEERING LOADS

Loads & structural response

Values identify their load case and whether they apply per component or to the whole rover.

Frame gravity solution is valid for its declared load case. Structural load factor: 1.5; structural safety factor: 2.

Quantity / scope	Prediction	Load case / interpretation
Required force per actuator `required_actuator_force`	2,500 N	Loaded raising/extraction envelope, including the model's breakout allowance.
Frame gravity reaction at each actuator `frame_gravity_actuator_force`	3,196 N	Mirrored loaded gravity sweep; includes the structural load factor. Separate from actuator sizing load.
Nominal force per wheel `frame_nominal_wheel_reaction`	360.7 N	Equal-share level-ground weight. Not a rock or slope wheel-load envelope.
Total cutting force `cut_force`	612.0 N	Across four drums at the selected bite and engagement.
Worst modeled contact pressure `ground_pressure`	18.8 kPa	Route pressure estimate, including load transfer and rocker factor.
Structural stress envelope `structure_stress`	98.2 MPa	Envelope of local member stresses; not one assembled-frame FEA result.
Structural deflection envelope `structure_deflection`	0.55 mm	Conservative combination of local/relative deflections.
Upper/lower mount relative displacement `frame_mount_relative_deflection`	0.17 mm	Frame gravity sweep.
Rocker-end flexure `rocker_deflection`	0.34 mm	Local beam-screen displacement.
Suspension-axle flexure `axle_deflection`	0.17 mm	Axle-overhang screen.
Arm-tip flexure `arm_deflection`	0.54 mm	Relative arm beam displacement.
Loaded CG height `loaded_cg_z`	425.1 mm	Fully loaded travel pose, above level ground.

APPENDIX / NATIVE UNITS

Design inputs & output data

144 inputs supplied by file; 0 CLI overrides; 5 remaining inputs use model defaults. Values below retain their native model units.

All effective inputs & model assumptions

Includes mission, soil, material properties and inactive legacy inputs. Source SHA-256: `48c31513a5397aee8c8978cc2095c0fdfae76ee34b4c5ff06f29bb10b75287dc`.

Input port	Effective value	Native units
lin_act_side_arm_offset	-0.0808	m
lin_act_top_arm_offset	0.0508	m
wheel_diameter	0.359	m
wheel_width	0.18	m
drum_diameter	0.359	m
drum_length	0.34	m
drum_rpm	120	rpm
feed_speed	0.2	m/s
dump_rpm	60	rpm
drum_gear_ratio	60	—
inlet_depth	0.025	m
arm_wall	0.0015875	m
battery_capacity	259.2	W*h
travel_actuator_length	0.433098	m
deposition_actuator_length	0.403098	m
actuator_retracted_length	0.407	m
actuator_extended_length	0.606	m
top_pivot_height	0.47	m
actuator_pivot_x	0.0681	m
bottom_pivot_height	0.26	m
arm_length	0.5188	m
pivot_x	0.1397	m
wheelbase	0.5	m
track_width	0.57	m
arm_box_width	0.0508	m
arm_box_height	0.0508	m
cross_tube_span	0.16	m
cross_tube_wall	0.003125	m
plate_spoke_width	0.01875	m
rocker_box_width	0.0508	m
rocker_box_height	0.0508	m
rocker_wall	0.0015875	m
axle_outer_diameter	0.033	m
axle_wall	0.001	m

Input port	Effective value	Native units
rocker_pivot_offset	0.0962	m
actuator_catalog_stroke	0.2032	m
minimum_ground_clearance	0.1524	m
clearance_radius_deflection_fraction	0.2	—
differential_bar_clearance_allowance	0.0254	m
empty_slope_degrees	30	deg
loaded_slope_degrees	20	deg
gravity	9.80665	m/s**2
run_time	900	s
traversal_time_one_way	45	s
travel_speed	1	m/s
acceleration	0.4	m/s**2
turn_angle	3.1415927	rad
overhead_time	8	s
electronics_power	25	W
actuator_idle_power	2	W
soil_density	1500	kg/m**3
berm_density	1500	kg/m**3
fill_fraction	0.9	—
delivery_efficiency	0.9	—
capture_fraction	0.65	—
cut_pressure	18000	Pa
internal_lift_factor	0.15	—
vertical_cut_fraction	0.5	—
force_imbalance	0.15	—
dump_revolutions	3	—
drum_efficiency	0.7	—
drum_gear_efficiency	0.8	—
drive_efficiency	0.72	—
drive_gear_efficiency	0.8	—
motor_rated_rpm	3000	rpm
traction_coefficient	0.6	—
rocker_load_factor	1.2	—
wheel_deflection_fraction	0.07	—
rolling_base	0.04	—
rolling_pressure_scale	200000	Pa
allowable_ground_pressure	25000	Pa
scrub_coefficient	0.25	—
turn_speed_fraction	0.35	—
turn_accel_fraction	0.5	—
chassis_height	0.35	m
drum_pair_gap	0.06	m
fixed_mass	6	kg
wheel_areal_mass	4	kg/m**2
drum_areal_mass	5	kg/m**2
motor_mass_per_watt	0.006	kg/W

Input port	Effective value	Native units
gear_mass_per_torque	0.05	kg/(N*m)
drive_fixed_mass	0.35	kg
drum_fixed_mass	0.5	kg
battery_specific_energy	140	W*h/kg
battery_usable_fraction	0.8	—
battery_c_rate	6.667	1/h
mass_score_weight	1	—
energy_score_weight	1	—
transport_clearance	0.08	m
mechanism_clearance	0.015	m
minimum_pivot_spacing	0.08	m
minimum_stroke	0.04	m
actuator_housing_allowance	0.04	m
pit_depth	0.45	m
actuator_efficiency	0.65	—
actuator_safety_factor	1	—
breakout_force	100	N
actuator_fixed_mass	0.6	kg
actuator_mass_per_force	0.0004	kg/N
actuator_mass_per_power	0.003	kg/W
actuator_mass_per_length	1	kg/m
stored_soil_density	1250	kg/m**3
usable_volume_fraction	0.9	—
scoops_per_revolution	4	—
max_dig_froude	0.3	—
max_dump_froude	0.2	—
drum_spinup_time	2	s
drum_torque_service_factor	1.25	—
minimum_dig_depth	0.005	m
minimum_payload	0.05	kg
fresh_site_distance	0.8	m
site_speed	0.1	m/s
available_excavation_area	12	m**2
tube_density	2700	kg/m**3
tube_youngs_modulus	6.9e+10	Pa
tube_yield_strength	2.4e+08	Pa
structure_safety_factor	2	—
structure_load_factor	1.5	—
allowable_deflection	0.005	m
joint_mass	2	kg
differential_mass	1	kg
obstacle_height	0.3	m
slope_angle	0.52359878	rad
frame_tube_diameter	0.038	m
frame_wall	0.003175	m
arm_tube_diameter	0.05	m

Input port	Effective value	Native units
plate_thickness	0.00635	m
cross_tube_inner_diameter	0.03175	m
wheel_rocker_gap	0.0127	m
plate_hub_radius	0.03	m
plate_cutout_clearance	0.005	m
frame_mount_clearance	0.003	m
minimum_axle_overhang	0.01	m
plate_buckling_k	1	—
cross_tube_density	2700	kg/m**3
cross_tube_youngs_modulus	6.9e+10	Pa
cross_tube_yield_strength	2.4e+08	Pa
drive_motor_power	835.4	W
drum_motor_power	835.4	W
drive_gear_ratio	100	—
motor_bus_voltage	12	V
x44_motor_mass	0.34019428	kg
drive_gearbox_mass	0.697	kg
drum_gearbox_mass	0.753	kg
drive_stator_current_limit	0	A
drum_stator_current_limit	0	A
actuator_force_rating	4003.3995	N
actuator_speed	0.009906	m/s
actuator_installation_allowance	0.1613408	m

Design variables and bounds

Optimizer search ranges and evaluated design variables. These ranges guide MDO and input validation; they are not all hardware requirements.

Port	Lower	Evaluated	Upper	Native units
lin_act_side_arm_offset	-0.3	-0.0808	-0.0508	m
lin_act_top_arm_offset	-0.15	0.0508	0.15	m
wheel_diameter	0.18	0.359	0.359	m
wheel_width	0.05	0.18	0.359	m
drum_diameter	0.16	0.359	0.359	m
drum_length	0.12	0.34	0.359	m
drum_rpm	5	120	120	rpm
feed_speed	0.0001	0.2	0.5	m/s
dump_rpm	5	60	60	rpm
drum_gear_ratio	9	60	125	—
inlet_depth	0.003	0.025	0.025	m
arm_wall	0.001	0.0015875	0.005	m
battery_capacity	10	259.2	300	W*h
travel_actuator_length	0.403098	0.433098	0.606298	m
deposition_actuator_length	0.403098	0.403098	0.606298	m
actuator_retracted_length	0.403098	0.407	0.606298	m
actuator_extended_length	0.403098	0.606	0.606298	m
top_pivot_height	0.35	0.47	0.7	m
actuator_pivot_x	0.02	0.0681	0.3	m
bottom_pivot_height	0.09	0.26	0.3	m
arm_length	0.3	0.5188	0.7	m
pivot_x	0.05	0.1397	0.4	m
wheelbase	0.38	0.5	0.65	m
track_width	0.5	0.57	0.699	m
arm_box_width	0.025	0.0508	0.1	m
arm_box_height	0.025	0.0508	0.1	m
cross_tube_span	0.16	0.16	0.46	m
cross_tube_wall	0.001	0.003125	0.006	m
plate_spoke_width	0.012	0.01875	0.07	m
rocker_box_width	0.02	0.0508	0.065	m
rocker_box_height	0.025	0.0508	0.09	m
rocker_wall	0.001	0.0015875	0.005	m
axle_outer_diameter	0.025	0.033	0.09	m
axle_wall	0.001	0.001	0.006	m
rocker_pivot_offset	0.02	0.0962	0.15	m
actuator_catalog_stroke	0.0508	0.2032	1.016	m

All output ports

Output port	Value	Native units
dry_mass	43.877166	kg
payload_mass	103.26206	kg
wheel_mass	6.4872883	kg
drum_mass	13.904561	kg
drive_mass	4.1487771	kg
battery_mass	1.8514286	kg
rover_length	1.4809588	m
rover_width	0.75	m
rover_height	0.74228231	m
ground_pressure	18757.686	Pa
actual_travel_speed	0.57462968	m/s
dig_time	31.522364	s
travel_time	90	s
dump_time	10	s
turn_time	4.308154	s
cycle_time	235.19479	s
cycle_energy	17.465433	W*h
cycles	3.8266153	—
berm_volume	0.2370865	m**3
run_energy	66.833493	W*h
average_power	267.33397	W
peak_power	3036.9492	W
mass_points	1585.0015	—
energy_points	354.74205	—
construction_points	1939.7436	—
no_load_speed	1.4582122	m/s
loaded_motor_speed_limit	1.1336356	m/s
loaded_power_speed_limit	8.4734395	m/s
stall_drive_force	7333.0043	N
actuation_time	67.320401	s
lowering_time	17.45427	s
arm_energy	9849.2695	J
arm_peak_power	336	W
deposition_reach_margin	0.1690325	m
deposition_position_margin	0.202902	m
deposition_clearance_margin	0.31068478	m
required_actuator_force	2500.0413	N
linkage_mass	5.9889015	kg
stroke	0.172902	m
stowed_x	0.56097938	m
stowed_y	0.56278231	m
travel_x	0.59403986	m
travel_y	0.51045704	m

Output port	Value	Native units
deposition_x	0.55555916	m
deposition_y	0.57018478	m
extended_x	0.60212291	m
extended_y	0.024801171	m
sweep_x_min	0.55555916	m
sweep_x_max	0.6585	m
actuator_leverage_max	3.6947203	—
actuator_attachment_distance	0.4409361	m
arm_physical_length	0.5188	m
arm_extension_mass	0	kg
actuator_body_margin	-0.1613408	m
chassis_clearance_margin	0.10745549	m
stowed_length	1.4809588	m
stowed_height	0.74228231	m
deployed_length	1.6759983	m
deployed_height	0.74968478	m
route_distance	18	m
route_turn_angle	360	deg
route_translation_time	64.857122	s
route_turn_time	25.142878	s
route_peak_acceleration	0.13207971	m/s**2
deposition_actuation_time	5.8403373	s
deposition_arm_energy	1708.9339	J
deposition_arm_peak_power	336	W
payload_volume	0.082609647	m**3
fresh_volume	0.1059098	m**3
storage_capacity	0.082609647	m**3
reachable_soil_mass	103.26206	kg
target_payload_mass	103.26206	kg
trench_depth	0.05	m
excavation_footprint	3.5249699	m**2
cut_force	612	N
drum_froude	2.890437	—
dig_overlap_time	0	s
dig_power	2675.9492	W
dump_power	124.36832	W
dig_energy	38783.122	J
dump_energy	493.55958	J
drum_power_margin	-817.2372	W
drum_torque_margin	-143.01613	N*m
dump_torque_margin	88.775371	N*m
dig_froude_margin	-2.590437	—
dump_froude_margin	-0.52260925	—
penetration_margin	124.28801	N
dig_traction_margin	-654.0006	N
cut_depth_margin	0.019875	m

Output port	Value	Native units
dig_reach_margin	0.029970808	m
payload_margin	103.21206	kg
structure_mass	5.4962096	kg
rocker_mass	0.90406009	kg
arm_linear_mass	0.84374831	kg/m
arm_envelope_radius	0.035921024	m
frame_section_margin	0.00725	m
arm_section_margin	0.0111125	m
structure_stress	98246985	Pa
structure_stress_margin	21753015	Pa
structure_deflection	0.00054665009	m
structure_deflection_margin	0.0044533499	m
structure_buckling_margin	2351.8759	N
empty_yaw_inertia	8.4755031	kg*m**2
empty_cg_x	-2.0242384e-17	m
empty_cg_z	0.36491307	m
loaded_yaw_inertia	50.871988	kg*m**2
loaded_cg_x	-2.4145252e-17	m
loaded_cg_z	0.42506465	m
half_yaw_inertia	29.673745	kg*m**2
half_cg_x	0.32113359	m
half_cg_z	0.41124761	m
loaded_sprung_mass	135.5991	kg
loaded_sprung_cg_x	0	m
loaded_sprung_cg_z	0.44564268	m
deposition_empty_yaw_inertia	7.7715907	kg*m**2
deposition_empty_cg_x	-2.530298e-17	m
deposition_empty_cg_z	0.38747083	m
deposition_loaded_yaw_inertia	45.600033	kg*m**2
deposition_loaded_cg_x	-2.4145252e-17	m
deposition_loaded_cg_z	0.47370824	m
deposition_half_yaw_inertia	26.685812	kg*m**2
deposition_half_cg_x	0.30033121	m
deposition_half_cg_z	0.45389919	m
relocation_time	18.203539	s
relocation_energy	246.58094	J
relocation_peak_power	19.841864	W
fresh_site_margin	0.06	m
site_drive_margin	172.91138	N
site_turn_time	9.9535391	s
mobility_energy	5914.2229	J
drive_peak_power	255.9662	W
turn_peak_power	127.47492	W
ground_pressure_margin	6242.3143	Pa
straight_motor_margin	3545.1805	N
turn_motor_margin	5327.1601	N

Output port	Value	Native units
straight_traction_margin	201.05347	N
turn_traction_margin	-250.36561	N
straight_power_margin	2494.7334	W
turn_power_margin	2587.2471	W
mobility_stability_margin	0.24427507	m
trench_traverse_margin	0.309	m
tip_stability_margin	-0.050331206	m
tip_worst_mask	3	—
obstacle_pose_margin	0.18273585	m
obstacle_tip_margin	0.15549579	m
obstacle_pose_valid	1	—
obstacle_evaluated_height	0.3	m
obstacle_worst_wheel	0	—
obstacle_chassis_pitch	0.33199714	rad
obstacle_chassis_roll	0.23027639	rad
obstacle_loaded_cg_x	-0.048811709	m
obstacle_loaded_cg_y	-0.032315749	m
obstacle_loaded_cg_z	0.47605622	m
obstacle_drum_wheel_gap_bound	0.048602307	m
obstacle_drum_wheel_margin	0.033602307	m
obstacle_clearance_pose_valid	1	—
obstacle_clearance_evaluated_height	0.3	m
slope_pitch_tip_margin	0.0045888083	m
slope_roll_tip_margin	0.039588808	m
slope_tip_margin	0.0045888083	m
slope_tip_angle	0.53165759	rad
rock_sweep_tip_margin	0.15549579	m
rock_sweep_worst_height	0.3	m
rock_sweep_raised_wheel	0	—
rock_sweep_pitch	0.33199714	rad
rock_sweep_roll	0.23027639	rad
rock_sweep_cg_x	-0.048811709	m
rock_sweep_cg_y	-0.032315749	m
rock_sweep_cg_z	0.47605622	m
rock_sweep_wheel_load_margin	111.5782	N
rock_sweep_load_height	0.3	m
rock_sweep_load_raised_wheel	1	—
rock_sweep_unloaded_wheel	1	—
rock_sweep_pose_valid	1	—
rock_sweep_evaluated_height	0.3	m
rock_sweep_samples	41	—
rock_sweep_force_balance_error	2.4868996e-13	N
empty_slope_slip	1	—
empty_slope_solution_valid	0	—
empty_slope_traversal_margin	-0.3437123	—
empty_slope_force_shortfall	32.020267	N

Output port	Value	Native units
empty_slope_sinkage	0.006051748	m
empty_slope_transfer_flag_count	0	—
loaded_slope_slip	1	—
loaded_slope_solution_valid	0	—
loaded_slope_traversal_margin	-0.12059223	—
loaded_slope_force_shortfall	40.878436	N
loaded_slope_sinkage	0.01447807	m
loaded_slope_transfer_flag_count	1	—
bottom_pivot_ground_clearance	0.1797	m
rocker_pivot_ground_clearance	0.1979	m
ground_clearance	0.1797	m
ground_clearance_margin	0.0273	m
bottom_pivot_axle_radius	0.019	m
rocker_pivot_axle_radius	0.0165	m
excavation_domain_margin	0.1545	m
rear_dig_depth	0.05	m
front_dig_x	0.64765983	m
front_dig_actuator_length	0.56838398	m
operating_reach_margin	0.029970808	m
cut_depth	0.025	m
feed_drive_distance	4.4044728	m
rear_dig_actuator_length	0.57602919	m
axle_ground_margin	-0.024801171	m
feed_per_scoop	0.025	m
front_dig_y	0.1545	m
rear_dig_y	0.1295	m
front_dig_depth	0.025	m
feed_margin	0	m
rear_dig_x	-0.6418187	m
feed_distance	3.1149942	m
inlet_width	0.34	m
feed_speed_limit	0.2	m/s
feed_time	23.522364	s
pair_cutting_time	16.574971	s
soil_supply_rate	0.0068	m**3/s
scoop_throughput	0.0068	m**3/s
feed_drive_energy	3077.3688	J
feed_drive_peak_power	210.3815	W
feed_motor_margin	5598.6782	N
feed_traction_margin	-654.0006	N
feed_power_margin	2527.5544	W
pit_depth_margin	0.29530117	m
transport_clearance_margin	0.30328231	m
wheel_first_margin	-0.028042956	m
startup_position_margin	-0.026098	m
dig_position_margin	0.199	m

Output port	Value	Native units
pivot_spacing_margin	0.14187059	m
pivot_mount_margin	0.239	m
pivot_longitudinal_margin	0.1103	m
bottom_mount_margin	0.0805	m
stroke_margin	0.132902	m
triangle_retracted_margin	0.1990325	m
triangle_extended_margin	0.041806688	m
toggle_margin	0.40085916	m
wheel_clearance_margin	0.0070029762	m
deployed_height_margin	1.7503152	m
actuator_pivot_longitudinal_margin	0.1819	m
actuator_toggle_margin	0.32684613	m
actuator_catalog_retracted_length	0.5644388	m
actuator_catalog_extended_length	0.7676388	m
actuator_unit_mass	2.5567141	kg
actuator_stroke_domain_margin	0.1524	m
actuator_force_margin	1503.3582	N
actuator_curve_margin	1503.3582	N
actuator_curve_valid	1	—
actuator_loaded_speed	0.010669057	m/s
actuator_raising_current	10.244796	A
actuator_lowering_current	14	A
actuator_raising_time	16.20593	s
plate_mass_total	0.91533317	kg
cross_tube_mass_total	0.59164044	kg
suspension_axle_mass	0.085175865	kg
suspension_bearing_span	0.3138	m
cross_tube_outer_diameter	0.038	m
plate_geometry_margin	0.033918366	m
frame_packing_margin	0.063725	m
plate_geometry_valid	1	—
structure_cg_z	0.28955335	m
structure_sprung_cg_z	0.30175016	m
structure_yaw_inertia	0.061217314	kg*m**2
plate_stress	12339294	Pa
plate_stress_margin	1.0766071e+08	Pa
plate_buckling_margin	2351.8759	N
cross_tube_bending_stress	46298132	Pa
cross_tube_bending_margin	73701868	Pa
frame_mount_relative_deflection	0.00016907674	m
rocker_deflection	0.00034366292	m
axle_deflection	0.00017010367	m
arm_deflection	0.00054495276	m
rocker_buckling_margin	149618.08	N
arm_buckling_margin	37178.143	N
frame_gravity_actuator_force	3195.8538	N

Output port	Value	Native units
frame_nominal_wheel_reaction	360.73572	N
frame_force_balance_error	9.094947e-13	N
frame_load_solution_valid	1	—